

4.1 Definitions

A graph is a finite nonempty set of vertices together with a set of edges, each edge being an unordered pair of distinct vertices. Two vertices joined by an edge are adjacent, and the edge is incident with each of them. A graph is a finite nonempty set of vertices together with a set of edges, each edge being an unordered pair of distinct vertices. Two vertices joined by an edge are adjacent, and the edge is incident with each of them. A graph is a finite nonempty set of vertices together with a set of edges, each edge being an unordered pair of distinct vertices. Two vertices joined by an edge are adjacent, and the edge is incident with each of them.

The degree of a vertex is the number of edges incident with it. Summing the degrees over all vertices counts every edge twice, which gives the handshake identity relating the degree sum to the number of edges. The degree of a vertex is the number of edges incident with it. Summing the degrees over all vertices counts every edge twice, which gives the handshake identity relating the degree sum to the number of edges. The degree of a vertex is the number of edges incident with it. Summing the degrees over all vertices counts every edge twice, which gives the handshake identity relating the degree sum to the number of edges.

A walk alternates vertices and edges; a path is a walk whose vertices are all distinct. A graph is connected when every pair of vertices is joined by a path, and a shortest such path realises the distance between them. A walk alternates vertices and edges; a path is a walk whose vertices are all distinct. A graph is connected when every pair of vertices is joined by a path, and a shortest such path realises the distance between them. A walk alternates vertices and edges; a path is a walk whose vertices are all distinct. A graph is connected when every pair of vertices is joined by a path, and a shortest such path realises the distance between them.